



Backchannel prediction for natural spoken dialog systems using general speaker and listener information

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Abstract

Backchannel responses are a crucial component of conversations enabling more effective communication through listener feedback. Current backchannel prediction models classify these responses into just three categories, using speech, text, and listener IDs. These IDs, which contain detailed personal information, cannot be applied in real-world dialog systems however, and three-category classification limits response generation capabilities. Therefore, we propose a model for predicting a backchannel's 'surface form' using only general speaker and listener embeddings. Our experiments show a 1.3% improvement in prediction accuracy when performing 3-category classification, and a 0.9% improvement when performing 11-category classification, compared to conventional ID embeddings, demonstrating an enhancement in performance that is deployable in real-world systems.

Index Terms: backchannel prediction, spoken dialog system, speaker and listener information

1. Introduction

Backchannels are conversational elements uttered by listeners during a speaker's speech to facilitate communication by providing feedback. These backchannels are not limited to short, verbal responses, but also include non-verbal cues such as nodding and smiling. These subtle signals (nods, non-verbal vocalizations, brief comments, etc.) not only convey understanding and empathy from the listener and provide feedback to the speaker, they also play a crucial role in sustaining the conversation. In recent years, as speech input and dialog applications based on spoken dialog systems have become more practical, achieving a system that can engage in dialog as naturally as humans still remains challenging. In order to realize dialog with AI agents comparable to human conversations, it will be necessary to replicate the natural backchannel responses which are frequently used during human interactions.

Previous research on backchannels has explored various aspects of this phenomenon, such as determining when dialog systems should produce backchannels [1–8]. Recent studies on classification of backchannels (BC) divide them into two main types: "Continuers" [9], which are generic BCs which acknowledge understanding and encourage the speaker to continue their narrative, and "Assessments" [10], which are specific BCs consisting of short, context-dependent responses [11–16]. Some studies have further split Assessment BCs into "Understanding" and "Empathetic" assessments, in order to make better predictions [17–19]. Other works have suggested annotating each function of the backchannel and classifying short feedback which is more detailed than a backchannel [20–22].

There have been two major challenges in recent research

on backchannel prediction. First, studies have demonstrated that incorporating additional information about speakers and listeners, alongside linguistic and acoustic features of their speech, can improve accuracy. However, when implementing a backchannel prediction model into a spoken dialog system, it is difficult to input personal IDs with extensive, personally identifiable information for many, or unknown, users. Therefore, these models need a way to acquire and utilize information about speakers and listeners that can be applied to any user. Second, predicting backchannel categories alone does not allow the system to determine which specific backchannel to use. Even within the "Continuer" category, there are multiple types of backchannels to choose from, and their nuances can vary subtly. Moreover, users can be put off if the system selects the same backchannel repeatedly. As a result, the model needs to be able to predict which specific backchannel the system should select, not just the correct backchannel category.

Therefore, in this paper we propose a backchannel prediction model that uses general information about speakers and listeners to predict the 'surface form' of backchannels, which is the more specific and appropriate type of backchannel response that should be selected by the system, taking into account the way backchannel prediction models are actually implemented in spoken dialog systems.

Major our contributions can be summarized as follows:

- Unlike previous research, which has proposed the use of inputs such as speaker IDs, which cannot realistically be used in actual spoken dialog systems, and backchannel category classifications such as Continuer or Assessment, we propose a model that uses self-reported, general speaker and listener information to predict the correct 'surface form' of backchannel responses, an approach which can be implemented in existing spoken dialog systems.
- By reproducing the model proposed by Ortega et al. in [13], we conduct a detailed comparison of different methods used to employ dialog participant information, as well as differences in performance across languages.
- Through experiments using dialog data, we analyze differences in the backchannel prediction accuracy of various models based on the presence or absence of speaker information, the methods used to employ this information, and the effect of speaker-listener relationships on backchannel selection.

2. Proposed Methods

The overall structure of our proposed model is shown in Fig. 1. We concatenate the features output from the language model, acoustic model, and speaker-listener model to obtain prediction probabilities for each backchannel through a classification

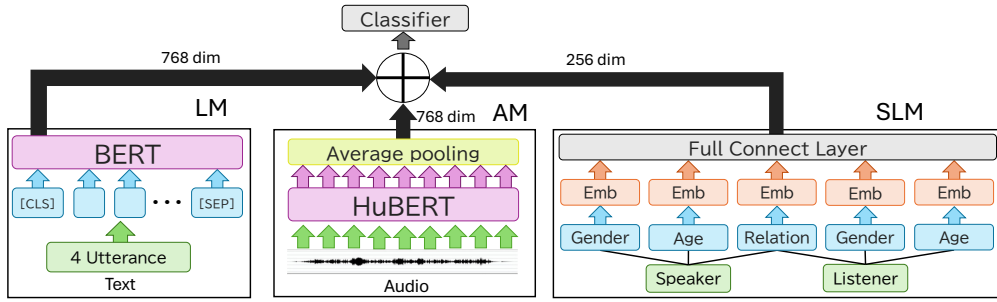


Figure 1: Structure of proposed model

Table 1: General information collected from speakers & listeners for CEJC data used in this study, with response totals

Gender		Age		Relationship	
Male	198	Less than 20	11	Family	80
Female	246	20 to 39	165	Friend	83
		40 to 59	157	Business	31
		60 or higher	101	School	9
				Other	14

layer. In the following, we will describe the details of each of these three model.

For our language model, we used BERT [23], which has been widely used in previous research. The input to the model consists of 4 turns of dialog text before the backchannel occurs, during which 768-dimensional features are obtained from the CLS (classification) token output.

For our acoustic model, we use HuBERT [24]. While Wav2Vec2.0 [25] has been used in predominantly previous studies, recent research has shown that HuBERT is more effective for classification prediction tasks [26]. The input to the model is 5 seconds of audio before the backchannel occurs. We obtained 768-dimensional features through average pooling of the HuBERT output.

The following information was used as input for the speaker and listener model; the speaker’s gender and age, the listener’s gender and age, and the category of the relationship between the speaker and listener (Family, Friend, Business, School, Other). The details of the input information are shown in Table 1. The model extracts 128-dimensional features for each piece of information through an embedding layer, concatenates these features, and processes them through a fully connected layer, ultimately obtaining 256-dimensional features.

3. Datasets

3.1. CEJC

The Corpus of Everyday Japanese Conversation (CEJC) [27] is a well-balanced, audio-visual corpus of typical, daily life Japanese conversations. It consists of 557 dialogs, with 200 hours of speech data in total, and the genders and ages of the speakers are balanced. The CEJC contains many multi-party (three or more speaker) conversations, however predicting backchannels under such condition is much more challenging. Therefore, in this study we only used the approximately 45 hours of conversations that involve only two speakers.

The Japanese backchannel responses obtained from the CEJC corpora which occurred frequently and were semantically classifiable were categorized into 11 ‘surface types’. The most common backchannel ‘surface form’ of each ‘surface type’, their English translations, and number of occurrences of each type of backchannel are shown in Table 2. Backchannels such as laughter are known as ‘non-verbal backchannels’ [28–30]. In previous studies, these responses have often not been con-

Table 2: Distribution of backchannels by ‘surface type’ categories in CEJC data used in this study, with examples

Backchannel	Total	Backchannel	Total
<i>Un</i> (Yeah)	8,745	<i>Ne-</i> (Isn’t it?)	407
<i>Unun</i> (Uh-huh)	2,680	<i>Fu-n</i> (Oh, really?)	605
<i>U-n</i> (Hmm)	3,304	<i>He-</i> (Wow)	526
<i>Ee</i> (I see)	572	<i>Ah</i> (Oh)	1,157
<i>Hai</i> (Yes)	2,258	(laughter)	3,582
<i>So</i> (Right)	886		

sidered to be backchannels, but since laughter tags are applied within the CEJC, it was included as one of the classification targets in this study. It should be noted that the backchannel “*Hai*” includes repetitions such as “*Hai hai*” and “*Hai hai hai*”, which are all treated as the same backchannel. Similarly, other backchannels that are semantically almost identical are also grouped together as the same backchannel ‘surface form’.

3.2. Data augmentation

The extracted backchannels are numerically imbalanced, so using all of the data as-is would result in accuracy bias. If we perform alignment using the less frequent backchannels, we can only use around 500 instances per type. In order to increase the amount of learnable data, we performed data augmentation. For audio data, we extracted 5-second segments from 0.2 and 0.5 seconds prior to the original segment and treated these as pseudo-correct data. For text data, we expanded the data by deleting the last 3 and 5 characters to correspond with the audio time segments. While the deletion of immediately preceding data does impact prediction accuracy, research by Adiba et al. [5] has demonstrated that reasonable prediction performance is still achievable under these conditions. This augmentation was made only for backchannels with less than 1,500 occurrences, and was not applied to backchannels with 1,500 or more occurrence. In the following experiments, the dataset is limited to a maximum of 500 samples per backchannel category without data augmentation, and a maximum of 1,500 samples per backchannel category with data augmentation.

4. Experimental Setup

4.1. Reproduction of previous research

To facilitate comparison with the results of previous research, backchannel responses extracted from the CEJC dataset were classified as either “Continuers”, “Assessments”, or “No-BC” (no backchannel) to obtain a three-category classification of the data. Following Ortega et al. [13], ‘uh-huh’ and its variants were classified as “Continuers”, while other backchannels were classified as “Assessments”. For the “No-BC” category, we also followed their methodology to extract utterances that were not considered backchannels. The number of instances within each classification were 5,000 for “Continuers”, 5,000 for “Assess-

Table 3: Comparison of accuracy and macro-average F1 scores for classification of 3 types of backchannel[%]

AM & LM	Dataset	SLM	Accuracy	F1 score
[13]	SWBD [33]	-	56.7	-
		[13]	58.9	-
[13]	CEJC	-	55.4	54.5
		ours	57.7	54.6
		[13]	57.0	55.8
Ours	CEJC	-	68.9	65.2
		ours	70.2	67.2
		[13]	71.9	69.1

ments”, and 10,000 for “No-BC”. This data was only used for the comparison experiment described in Section 5.1.

To compare the effectiveness of our approach with conventional listener-only information embeddings, we conducted a comparative analysis based on Ortega et al. [13]. Following their methodology, linguistic features were extracted using the five words preceding a backchannel event, as encoded by Word2Vec [31]. Acoustic features were derived from the 2 seconds preceding the backchannel event, using 13-dimensional MFCCs extracted using the OpenSMILE toolkit [32], and processed through a Convolutional Neural Network (CNN). Listeners were assigned unique IDs in numerical order, and these IDs were processed through an embedding layer to generate listener-specific features. Using the CEJC dataset, we replicated the model proposed in [13] to facilitate a direct comparison with our proposed approach. Additionally, to evaluate the difference in performance between listener ID embedding as used in [13] and our proposed use of general speaker-listener information, we conducted experiments where we replaced the speaker-listener model in our proposed architecture with listener ID embeddings.

4.2. Training settings

The pre-trained models used in this study were independently developed by Inui Lab at Tohoku University¹ and by Rinna². While some previous studies have relied on the output of these pre-trained models, we performed fine-tuning to adapt the models to our task. The training dataset consisted of up to 1,500 samples from each backchannel category, with the data split into training, validation, and test sets a ratio of 6:2:2.

The classifier architecture comprised three layers, with hidden layer sizes of 1,024 and 512, followed by the number of output classes. Dropout layers with a dropout rate of 0.25 and ReLU activation functions were applied between each layer. The model was trained for 100 epochs using a batch size of 128. We employed the Adam optimizer with a learning rate of 1e-5 and used cross-entropy loss to guide the learning process.

5. Results

5.1. Comparison with previous studies

Table 3 shows the three-category backchannel classification results for the pre-trained base model only, the conventional classification model proposed by Ortega et al. in [13], and for our proposed model. The upper row of the table shows the accuracy rates reported in [13], where they achieved a 2.2% improvement in accuracy through the implementation of listener ID embeddings. The middle row displays results for the pre-

¹<https://github.com/cl-tohoku/bert-japanese>

²<https://huggingface.co/rinna/japanese-hubert-base>

Table 4: Accuracy and macro-average F1 scores for classification of 11 types of backchannels various conditions [%]

LM	AM	DataAug	SLM	Accuracy	F1 score
+	-	-	-	31.1	29.2
+	-	+	-	35.7	34.2
-	+	-	-	26.3	22.6
-	+	+	-	29.1	24.8
+	+	-	-	36.5	35.0
+	+	+	-	37.4	37.1
+	+	+	ours	39.2	38.7
+	+	+	[13]	40.3	39.9

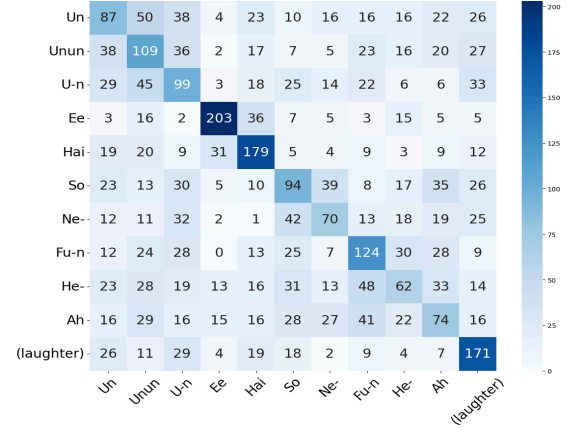


Figure 2: Confusion matrix for classification of test data using our proposed method

trained base model only, our replication of the model proposed in [13] trained with the CEJC dataset, and for our model, to compare the results when using listener ID embeddings versus our proposed speaker-listener model. While listener ID embeddings yielded improvements of 1.6% in accuracy and 1.3% in the F1 score, our proposed speaker-listener model achieved a 2.3% improvement in accuracy and a slight 0.1% improvement in the F1 score. The bottom row presents results when using HuBERT acoustic model and BERT language model with the pre-trained only, conventional, and proposed backchannel classification models. Here, the model using listener ID embeddings demonstrated improvements of 3.0% in accuracy and 3.9% in the F1 score, while our proposed speaker-listener model showed improvements of 1.3% in accuracy and 2.0% in the F1 score.

5.2. Main results

Prediction results when classifying 11 types of backchannels are presented in Table 4. Looking at the upper portion of the table, we can see that data augmentation improved accuracy rates and F1 scores across language models, acoustic models, and combined language-acoustic models when using the pre-trained model only. Examining the lower portion of the table, our proposed approach incorporating language, acoustic, and speaker-listener models demonstrated improvements of 1.8% in accuracy and 1.6% in F1 score through the integration of the speaker-listener model. However, in comparison to the conventional method using listener ID embeddings, our proposed model scored 0.9% lower in accuracy and 1.2% lower in F1 score. Fig. 2 shows a confusion matrix of the classification result for our proposed model when using the test data, corresponding to the parameters of the seventh row in Table 4. The matrix reveals notably higher accuracy rates for “Ee”, “Hai”, and “(laughter)” compared to the other backchannel responses.

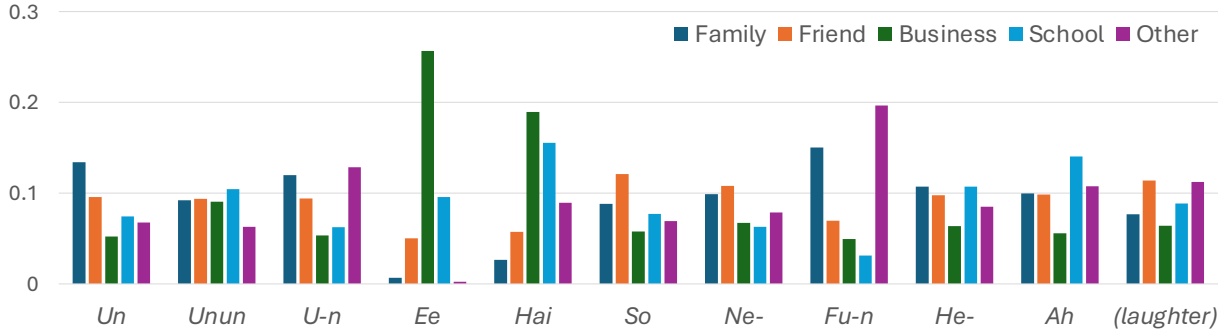


Figure 3: Percentage of each speaker-listener relationship for each backchannel response

6. Analysis and Discussion

6.1. Analysis

Regarding the results of the main experiment this study, high accuracy rates were observed for the backchannel responses “Ee” (I see), “Hai” (Yes), and “(laughter)”. “Ee” and “Hai” are honorific expressions in Japanese, typically employed when addressing individuals of higher social status. The superior classification performance for these response suggests that our model effectively incorporates speaker and listener relationship information.

Fig 3 shows the distribution of the five speaker-listener relationship types used as input to model for each of the 11 backchannel responses. We can see that the backchannels “Ee” and “Hai” occur more frequently in Business contexts compared to Family or Friend situations. This distribution aligns with their function as polite backchannel expressions which are more frequently used in professional relationships than in more familiar ones. This clear contextual pattern likely contributed to the model’s ability to predict these expressions more accurately, using the provided speaker-listener relationship information. While “(laughter)” shows no notable distribution patterns across relationship types, its high prediction accuracy can be attributed to its contextual nature, since listeners often respond with laughter when the speaker has just laughed. This makes laughter relatively easy to predict, based on dialog history and acoustic features.

6.2. Discussion

Our experiments demonstrated improvement in backchannel classification accuracy through the incorporation of general speaker-listener information in both three-category backchannel prediction (for comparison with previous studies) and in the 11 category backchannel prediction task used in this study. However, the accuracy of the proposed method was slightly lower than that achieved by methods proposed in previous research which use personal ID embeddings. It is important to note that while methods using personal ID embeddings have achieved higher accuracy, they face a significant limitation: They cannot be practically implemented in real-world spoken dialog systems due to the difficulty in collecting and inputting personal ID profiles for random users. In contrast, our approach, which relies on easily obtainable user inputs such as gender and age, can be utilized by any user. Thus, our research highlights the trade-offs between performance and practicality when using systems with and without detailed speaker-listener information, especially in relation to real-world deployment in spoken dialog systems.

While our analysis focused on the relationship between backchannel responses and speaker-listener information, other factors warrant investigation. For example, Figueroa et al [21]

have demonstrated that backchannel types vary according to the speaker’s gender. Additionally, differences based on whether conversation participants are in the same age group, or belong to different generations, should also be examined. Furthermore, Park et al [19] have shown that incorporating factors such as elapsed time since the beginning of the dialog, as well as conversation topics, can improve prediction accuracy. We plan to investigate these elements, along with contextual factors such as conversation location.

7. Conclusion

In this study we have proposed a model that predicts 11 types of backchannel ‘surface forms’ using general speaker and listener information such as gender, age, and relationship. When used for three-category backchannel prediction, as in previous studies, our proposed approach using speaker-listener information showed a 1.3% improvement in classification accuracy. However, this was 1.7 % lower than the accuracy of a conventional method using personal ID embeddings. When used for the prediction of 11 types of backchannels, our proposed method demonstrated a 1.8% improvement in accuracy compared to using only a pre-trained model, however this was still 0.9% below the accuracy achieved by the conventional method using personal ID embeddings. Although we did not exceed the accuracy achieved in previous research, we have established a new benchmark for backchannel prediction accuracy using more generalized speaker and listener information, an approach which takes into account the issue of practical implementation in spoken dialog system.

8. Limitation

In this study we have successfully validated the effectiveness of utilizing general speaker-listener information as opposed to previously employed individual ID-based approaches. However, when applied to a task involving the prediction of 11 types of ‘surface form’ backchannel responses, our proposed method achieved a maximum accuracy of just over 39%, which while notably low, was only 0.9% lower than a conventional method. This limited performance can be attributed to the fundamental difficulty of predicting backchannels in conversation, due to the lack of strict grammatical or ‘naturalness’ rules limiting possible backchannel responses in given contexts. Moving forward, we propose transitioning from the current single-label learning approach to a multi-label learning framework that can accommodate multiple valid responses. It is important to note that our current dataset does not include multi-label annotations. Therefore, careful consideration must be given to both the methodology for label annotation and the evaluation metrics for such a model.

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10. References

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